Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Full-stack software engineer with 2+ years building secure, maintainable web applications using Python, TypeScript, React, SQL, and REST APIs. Experienced in legacy modernization, automated testing, CI/CD, Azure, Agile delivery, debugging, and AI-assisted development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built secure Python APIs and Azure SQL services for 500+ users, optimizing indexed queries, schemas, and execution paths to reduce response latency from 1.2 seconds to 350 milliseconds.
- Refactored a legacy job orchestration workflow into maintainable components with explicit states and failure isolation, tripling throughput while improving testability, observability, debugging, recovery, and overall production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Python and Flask REST services handling 10,000+ daily requests, implementing secure authentication, validation, persistence, and error handling while troubleshooting production issues affecting more than 150 users.
- Delivered React workflows and multithreaded processing for 1,000+ files with AWS RDS persistence, automated validation, progress monitoring, and failure recovery, reducing turnaround 60% under concurrent workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built maintainable Python and PyTorch pipelines processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, automated validation, performance profiling, technical documentation, and systematic debugging.
- Developed modular data-processing components combining map geometry and agent context, improving stability 15% while documenting technical specifications and validating behavior across diverse simulated operating scenarios and workloads.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services and SQL integrations using AWS Glue, S3, filtering, pagination, and caching, improving production response times 30% and processing efficiency 68% across connected business applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating defect investigation, code review, regression verification, and reliable CI/CD releases across multiple environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed TypeScript and Node.js services across eight microservices with authentication, role-based access, WebSocket messaging, and synchronization, securely processing more than two million Azure SQL records through documented APIs.
- Automated Docker builds, functional tests, staged deployments, and monitoring through Jenkins and Azure, reducing release time 50% while improving code quality, deployment consistency, and overall operational visibility.

Skills

Languages & Web: Python, TypeScript, JavaScript, SQL, Java, React, Node.js, Flask, FastAPI, HTML, CSS, REST APIs, WebSockets, client-server architecture
Data & Cloud: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, AWS RDS, relational databases, schema design, query optimization, caching, AWS, Azure
Quality & Security: Automated testing, unit testing, integration testing, PyTest, Mockito, authentication, role-based access, secure APIs, debugging, logging, monitoring, failure recovery
Delivery & Collaboration: Git, GitHub Actions, Jenkins, CI/CD, Docker, Kubernetes, Jira, Agile, code reviews, requirements analysis, technical documentation, production support, AI-assisted workflows

Projects

Big-Brain | Live | *Python, React, PostgreSQL, GCP, REST APIs, AI*
- Built a secure AI-assisted document application over 50,000+ files using React, Python APIs, PostgreSQL, semantic retrieval, optimized indexing, automated validation, and cloud deployment with sub-second search.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built a modular TypeScript and Java database tool supporting 10+ engines with schema-aware SQL generation, query validation, metadata exploration, error handling, automated testing, and Docker-backed CI/CD deployment.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a client-server platform supporting 50,000+ users and one million transactions, integrating secure REST APIs, WebSockets, MongoDB persistence, Kubernetes deployment, production monitoring, and automated release workflows.